Coordinate Transformations

ECE/MAE 5930 Mobile Robotics

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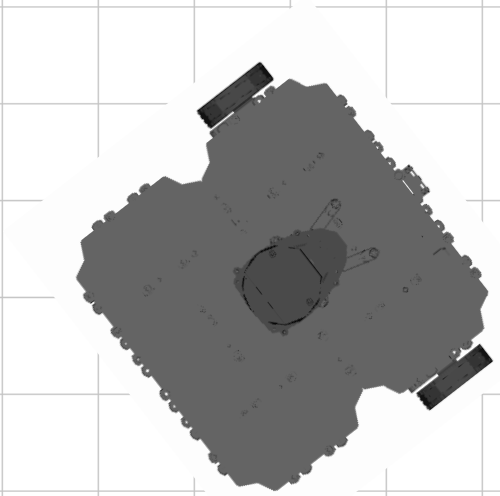
Motivation

- Previously
  - Discrete planning: graph methods using vehicle as a point and global coordinate frame
  - Polygon representation: use polygons, set operations, and predicates to represent entire body or world (i.e., no need for point assumption)
- Coordinate frames and transformations
  - It is convenient to express information in different frames of reference
  - It is important to be able to transform data to different reference frames
- Rigid body transforms
  - Two dimensional rotation
  - Two dimensional rotation + translation
  - Three dimensional rotation
  - Three dimensional rotation + translation



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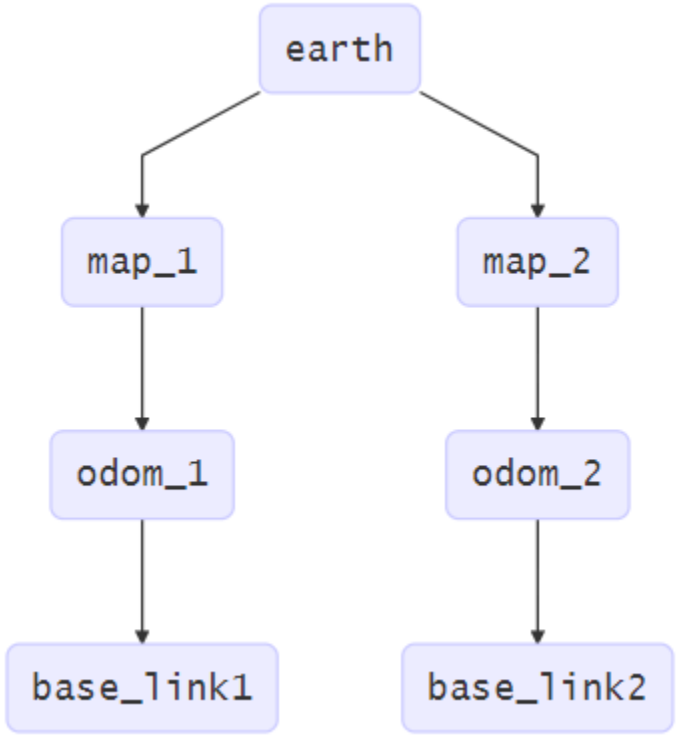
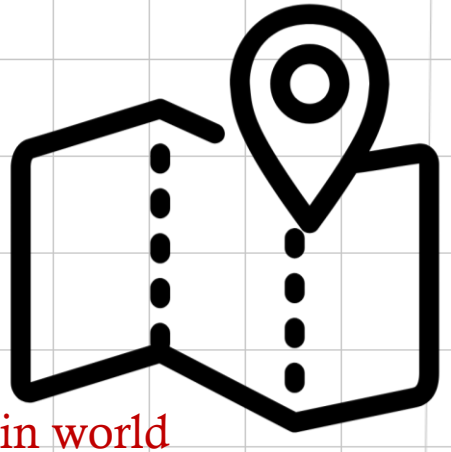
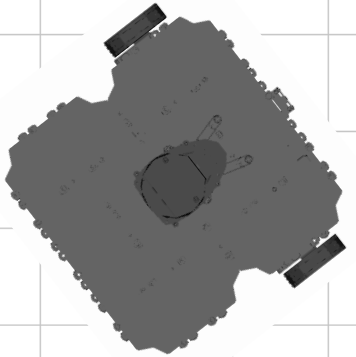
Reference Frames

- To define "where" a point is, we need a point of reference
- ROS uses a host of frames: Two robot example



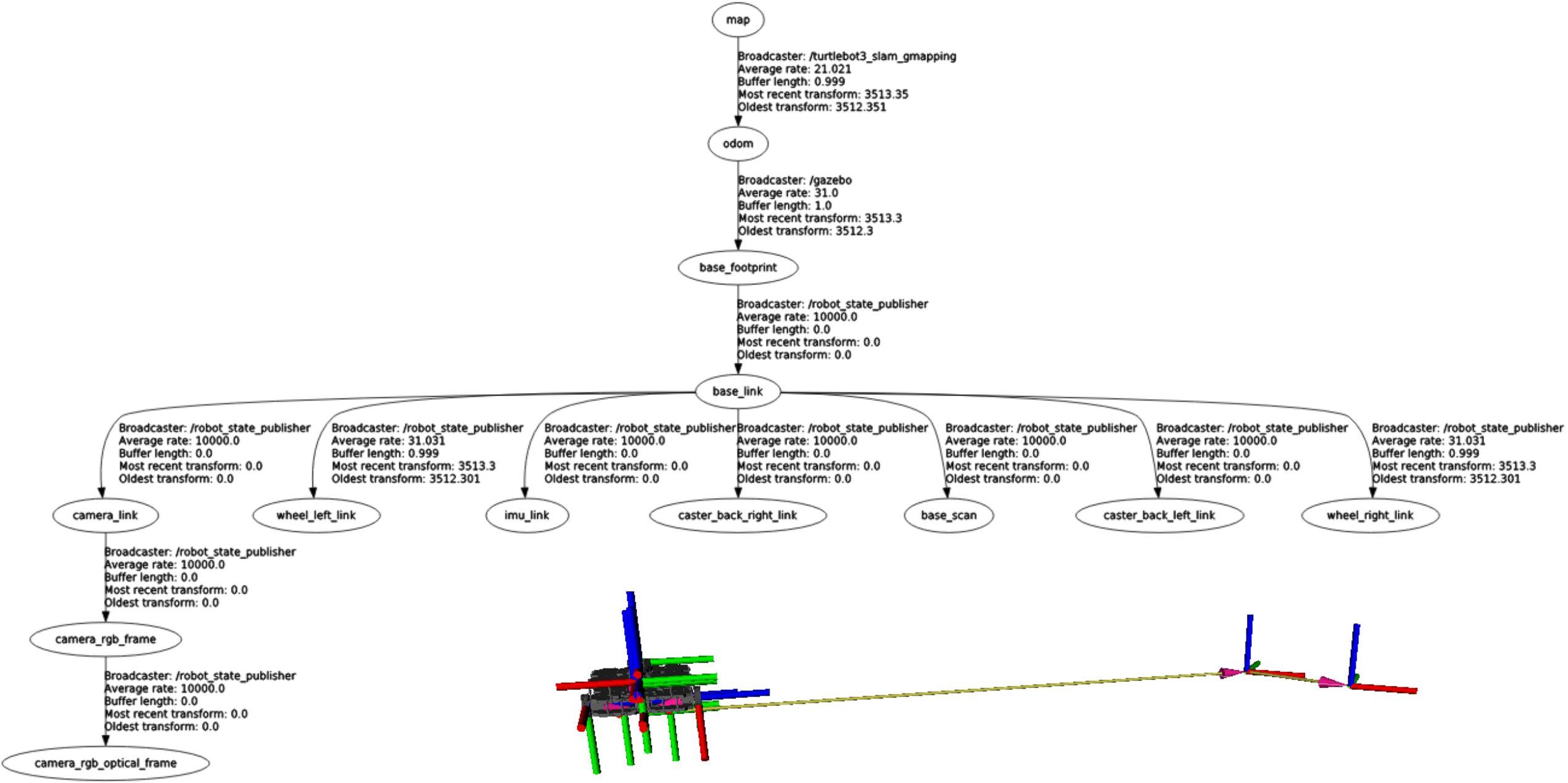


Earth: GPS – WGS-84
Map: Some fixed point and orientation in world
Odom: Fixed point and orientation – based on where the vehicle starts
Base_link: frame on robot



http://www.ros.org/reps/rep-0105.html

Example of our Simulator



Notation

- Same points can be described in any of the reference frames
- Pre-subscript – denote the frame

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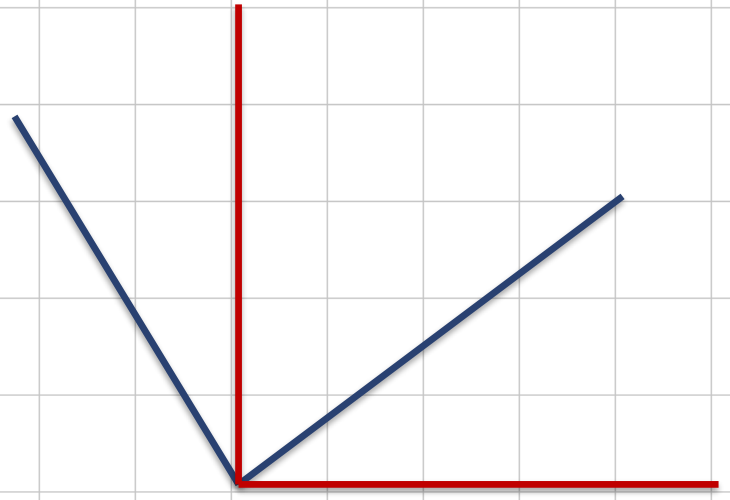
1st Transform: 2D Rotation

- Origin points of frames are co-located
- LaValle describes counter-clockwise rotation – this is just a rotation about z-axis
- Recall direction cosine matrices
  - Just describe the elementary vectors



- Can transform point between frames
- Now transform it back

- Example

- What is the transform matrix for transforming to a new frame and then transforming back?

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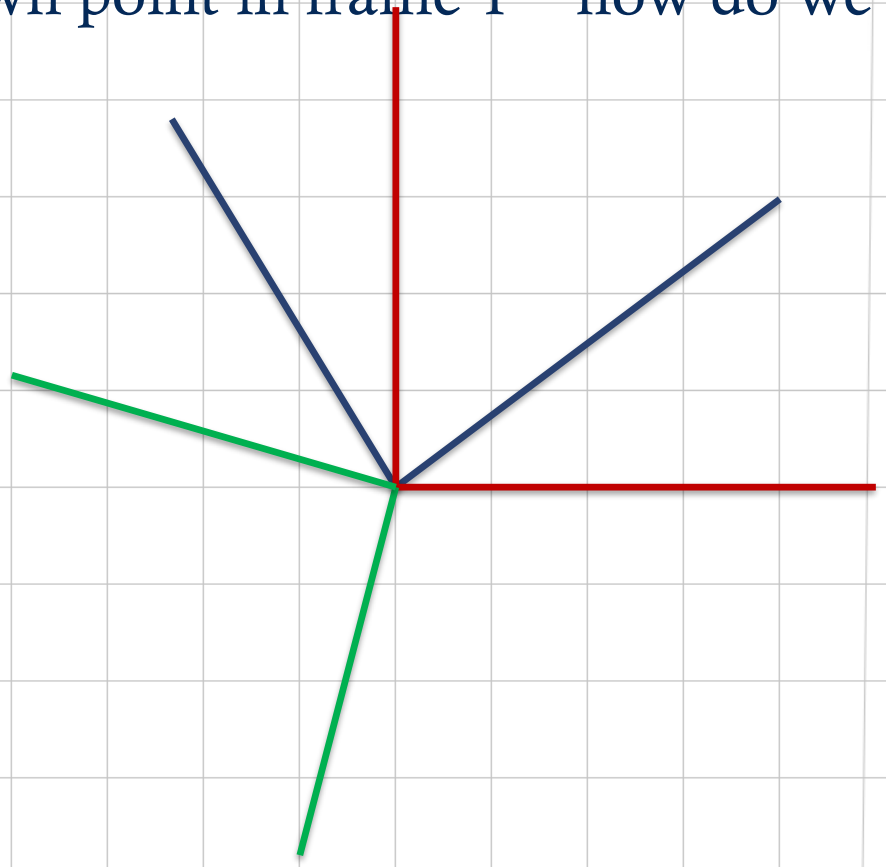
2D Rotation: Two Rotations

- We know the two rotation angles and have a known point in frame 1 – how do we get it to frame 3?



- Facts
  - 2D rotation matrices are commutative (3D are not)
  - We defined a point-fixed rotation (i.e. the point stays put), a fixed-frame rotation is the inverse (i.e. the frame stays put and the point moves)
  - Rotation matrices belong to the Special Orthogonal group of two dimensions (SO(2))
- Recall the properties of a group
  - Closure under group operator
  - An identity element
  - Inverse element
  - Associativity

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2nd Transformation – 2D Rotation and Translation

- This group of transforms are known as the Special Euclidean group of two dimensions

- Say we have a point and want to rotate and translate:

Chaining 2D Transformations

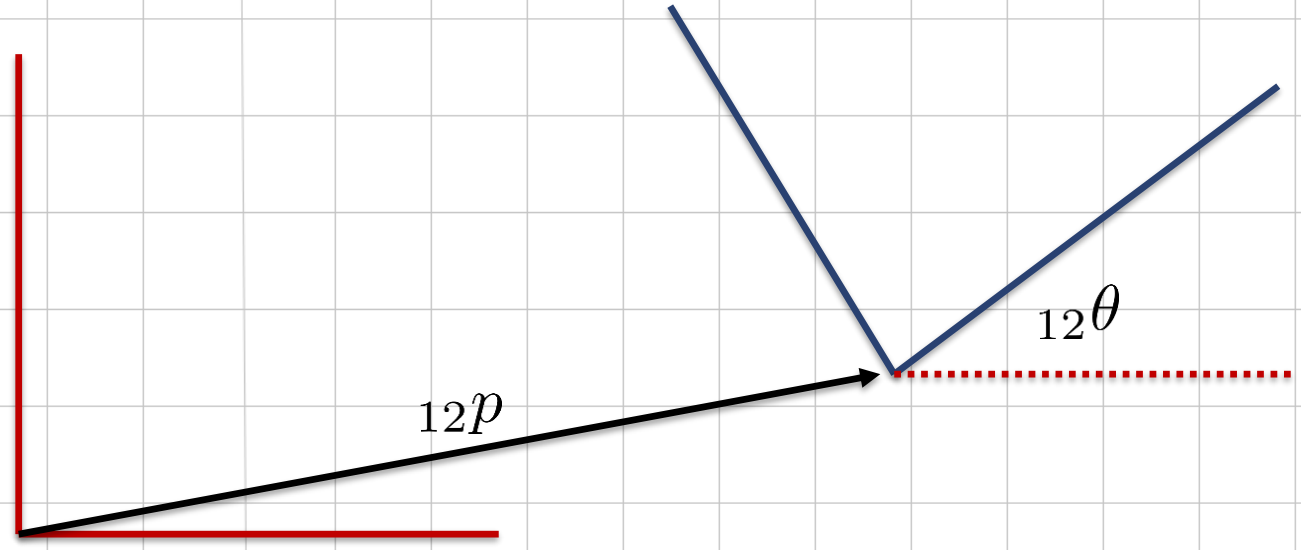
- Say we know the relationship between frames 1-2 and frames 2-3, how do you express a point known in frame 3 coordinates in frame 1 coordinates?

Taking the Inverse

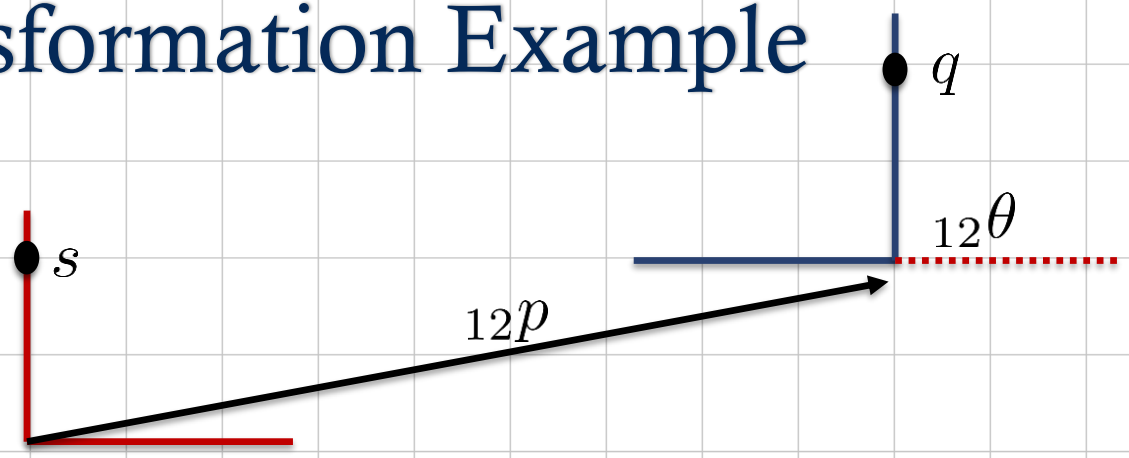
- We said that rotation+translation belongs to a group, then an inverse must exist

$$_1q = {}_{12}p + {}_{12}R \cdot {}_2q$$

$$_2q = {}_{21}p + {}_{21}R \cdot {}_1q$$



A Simple Transformation Example



- What are the transform variables?
- Express q in frame 1 coordinates
- Find the inverse transform parameters
- Express s in frame 2 coordinates

3rd Transformation – 3D Rotation

- Recall direction cosine matrices

- Unlike 2D scenario, there is not a single parameter – there are three
- Elementary direction cosine matrices
  - Lavelle talks about clockwise and counter-clockwise rotations
    - Confusing – Use positive right-hand rotations about an axis

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3D Rotation – Elementary Matrices

Fact: Any 3D rotation can be expressed using the Elementary direction cosines

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3D Rotations – Euler Coordinates

- Fact: Any 3D rotation can be expressed using the Elementary Direction Cosine matrices (i.e. they form a basis)

- Example: Euler X-Y-Z (roll, pitch, yaw)

$$R(\gamma,\beta,\alpha) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}$$

  - Which rotation is performed first?
  - Extracting roll-pitch-yaw

Note error in book

$$\begin{cases} \alpha = atan2(r_{21}, r_{11}) \\ \beta = atan2(-r_{31}, \sqrt{r_{32}^2 + r_{33}^2}) \\ \gamma = atan2(r_{32}, r_{33}) \end{cases}$$

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

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3D Rotation Properties

- 3D Rotations are **<u>not commutative</u>**
- 3D Rotations belong to the Special Orthogonal group of three dimensions (SO(2))
  - All group properties expressed for 2D are same for 3D rotations
    (Group operator, identity element, inverse element, associativity)
- The 3D inverse is same as 2D inverse
- 3D rotation matrices can be chained in the same fashion as 2D rotation matrices
- Take extreme care when you are given Euler angles as there are 12 possible combinations for Euler angles
- Major differences between 2D and 3D rotations:
  1. 2D has one angle (one degree of freedom) and 3D has three degrees of freedom
  2. 3D is not commutative

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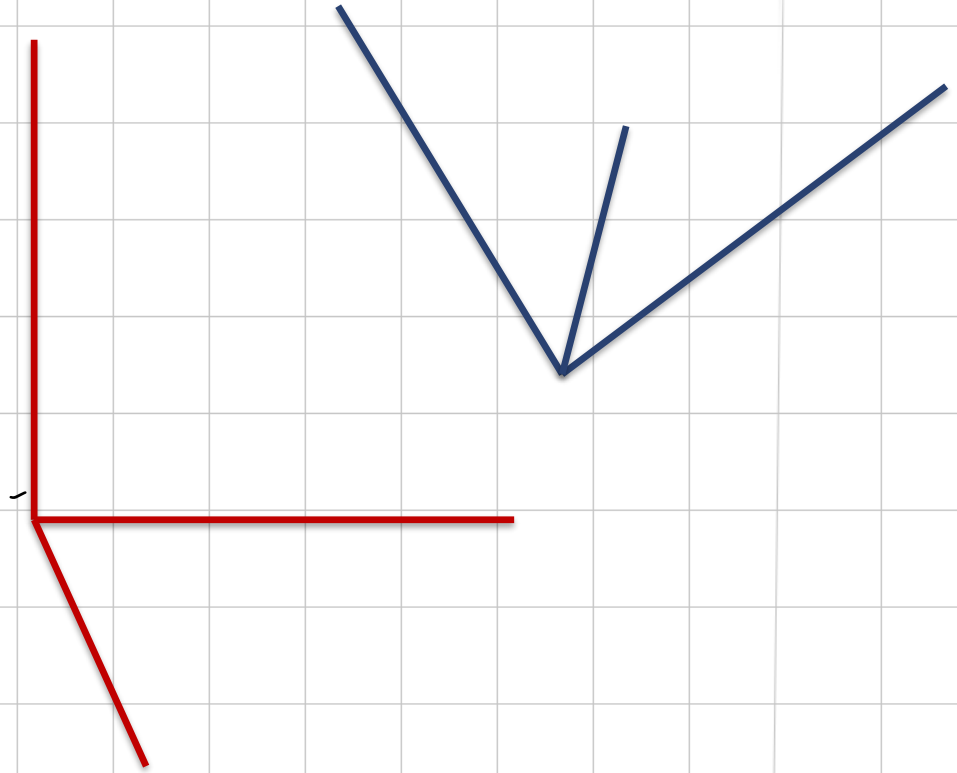
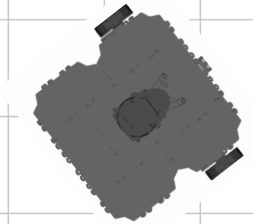
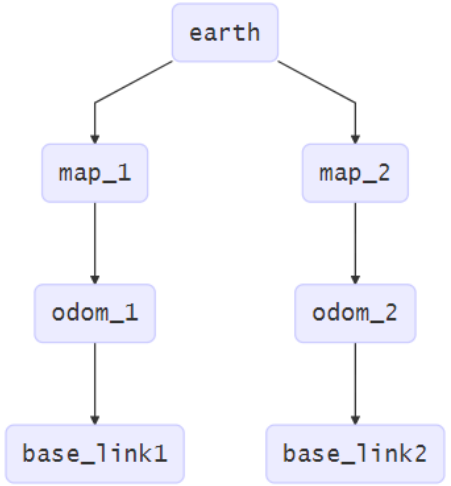
4th Transform – 3D Rotations & Translations

- This group of transforms are known as the Special Euclidean group of three dimensions

- Say we have a point and want to rotate and translate:



Summary

- Frames of reference are ubiquitous to sensory and position information
- Discussed four transforms
  - 2D rotation (SO(2))
    - 2x2 matrix representation with one degree of freedom
    - Group properties and group operator
    - Inverse = Matrix transpose
    - Commutative
  - 2D rotation and translation (SE(2))
    - 2x2 matrix and 2x1 vector – total of 3 degrees of freedom
    - Held off on group properties
    - Inverse – slightly more complicated, but still fairly simple
  - 3D rotation (SO(3))
    - 3x3 matrix with 3 degrees of freedom
    - Not commutative, otherwise same as 2D rotation
    - Elementary rotations and Euler angles
  - 3D rotation and 3D translation (SE(3))
    - 3x3 matrix and 3x1 vector – total of 6 degrees of freedom
    - Operations same as SE(2)